

Experiment 4: Current Behavior in an RL circuit

1. Objective:

- 1) To observe the rise and decay characteristic of an inductor using an oscilloscope.
- 2) To verify the time constant of an RL circuit.

2. Background:

An inductor is a passive electronic component that stores energy in the form of a magnetic field. In its simplest form, an inductor consists of a wire loop or coil. If a source emf is introduced in a circuit containing an inductor and a resistor, an induced emf ($\mathcal{E}_L$) will be observed across the inductor while there is a change in the current (like switch 'on' or 'off'), written below:

$$\mathcal{E}_L = -L \frac{di}{dt} \quad (1)$$

It follows the **Faraday's law of induction**. The "minus" sign in the equation 1 indicates that $\mathcal{E}_L$ opposes the change in current, explained by the **Lenz's Law**.

In this experiment we will observe the decay (positive to zero and negative to zero) of voltage across the inductor in an exponential fashion, like we observed in the *RC* circuit experiment. However, there is no such terms as "charging" and "discharging" in an inductor.

If a voltage (square wave) V_O is introduced in a circuit containing R and L , when the current rises exponentially to $\frac{V_O}{R}$, and because of this change the voltage across the inductor will decay exponentially (positive to zero) (see equation 1), written as,

$$V_L(t) = V_O e^{-t/\tau}. \quad (2)$$

Where τ_L is the inductive time constant of the inductor measured in terms of second, given by

$$\tau = L/R. \quad (3)$$

Similarly, if the constant emf V_O is withdrawn, current decays exponentially and the voltage across the inductor also decay exponentially (negative to zero), written as,

$$V_L(t) = -V_O e^{-t/\tau}. \quad (4)$$

In figure 2 both current and voltage waveforms are given schematically with respect to time.

Here the inductive time constant, τ is the time required to decay the inductor, through the resistor, 37% of its initial voltage.

For this experiment we will use the circuit in figure 1.

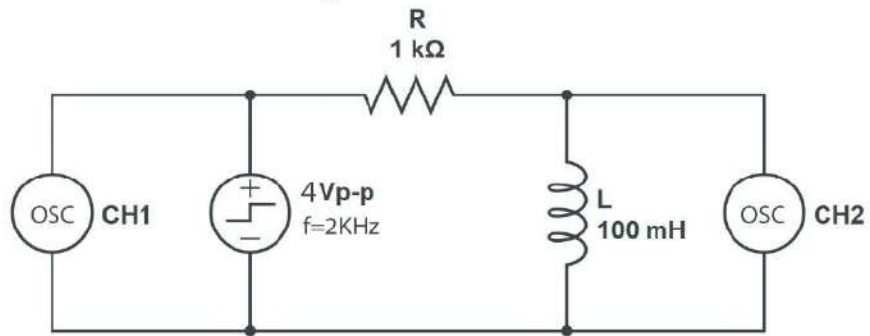


Figure 1: RL Series Circuit

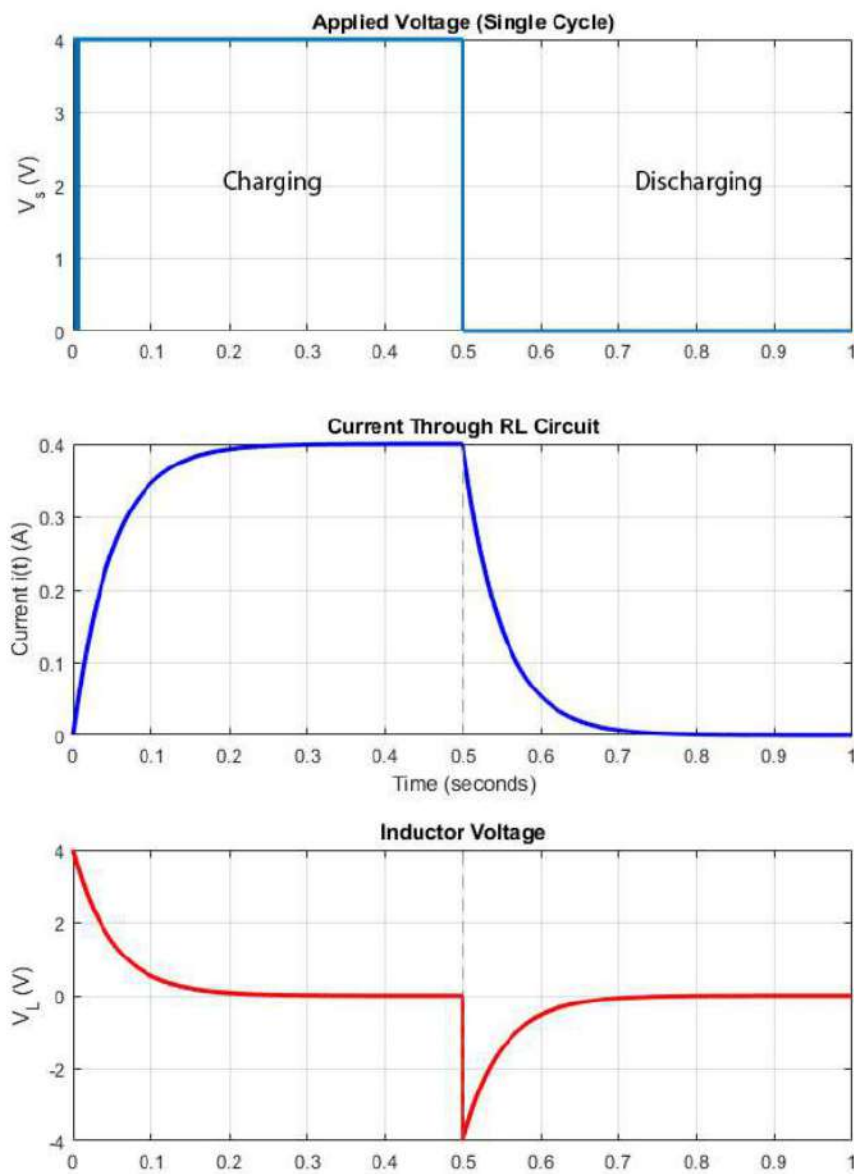


Figure 2: Current and voltage waveform with respect to time

3. Procedures and Observations:

- a) Measure the value of the resistor, $R_{measured} =$
- b) Connect R and L in series with the AC signal generator. (For reference, please see the circuit given above in figure 1).
- c) Connect the oscilloscope's channel 1 to the signal generator, choose **square wave** as an input signal and observe the signal on the oscilloscope.
- d) Make sure that the channel is set to **DC coupling**.
- e) Measure the amplitude and period of the input signal and make sure that the amplitude is set to $4V_{p-p}$ and the frequency to 2 kHz with 50% duty cycle.
- f) Use the offset knob to raise the signal by 2V so that the base of the signal is set at 0 V.
- g) Connect the oscilloscope's channel 2 across the inductor and observe the output.
- h) Measure the voltage across the inductor at 20 μs (approximately) interval and record the data in the following table.
- i) For calculation of inductor voltage during decay, measure the maximum inductor voltage during charging and use it as V_o in Eqn. 1.
- j) Measure the inductor voltage and record in the following table.

Lab Report:

Date:	
Name of the Students and IDs	(1)
	(2)
	(3)

Data Tables:

Table 1: Time dependent for the voltage decay (**positive to zero**) of an inductor

[illegible]Table 2: Time dependent for the voltage decay (**negative to zero**) of an inductor[illegible]

Tasks and Questions:

#1: Use data obtained in Table 1 to plot inductor voltage (V_L) vs time (t) graph. Calculate the time constant and compare with the theoretical value. Label the axes properly.

Time constant (theoretical) =

Time constant (measured from the graph) =

% of error =

#2: Use data obtained in Table 2 to plot inductor voltage (V_L) vs time (t) graph. Calculate the time constant and compare with the theoretical value. Label the axes properly.

Time constant (theoretical) =

Time constant (measured from the graph) =

% of error =

Results:

Discussion: